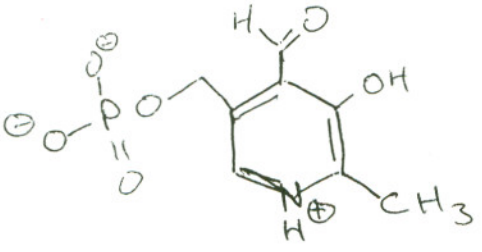
Biochemistry Cumulative

September 19, 1992

1. Recently Dr. Barry Hurlburt gave an excellent biochemistry seminar which only one graduate student attended, in spite of three days advance notice and an announcement three hours before the seminar. Name the protein system he works with. (5 points) Give a one or two sentence description of some experiment he performed and what he learned. (5 points) (This time, 10 points; next time, the whole exam.)

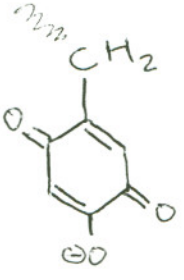
2. (30 points) Recently Dr. Judith Klinman also gave an excellent biochemistry seminar. Hopefully you made time in your busy schedule of experiments to attend this seminar. She spoke on the mechanism of copper amine oxidase. This enzyme catalyzes the reaction: $RCH_2NH_2 + O_2 + H_2O \rightarrow RCHO + NH_3 + H_2O_2$. For many years it was thought that the cofactor of this enzyme was pyridoxal phosphate, shown below. Show how pyridoxal phosphate could catalyze a deamination reaction, i.e. write a mechanism (you need not worry about the oxidation state of the cofactor).

pyridoxal phosphate



3. (30 points) Pyridoxal phosphate was shown not to be the cofactor, rather it was TOPA, shown below. Show a mechanism for the catalysis of the above reaction using TOPA. This time include a mechanism to return the cofactor to its original oxidation state.

TOPA



4. (20 points) A key experiment in showing pyridoxal phosphate could not be the cofactor was the treatment of the enzyme:substrate complex with a radiolabeled reducing agent. If you were doing this experiment what reducing agent would you use and why? Show the intermediate that this reducing agent would act on if pyridoxal phosphate were the cofactor and the expected product of the reduction. Show the intermediate that the reducing agent would act on if TOPA were the cofactor and the expected product of the reduction. What distinguishing characteristic of the two products was used to show that pyridoxal phosphate could not be the cofactor? Give a mechanistic based justification based on the products.

5. (10 points) Briefly describe the biogenesis of TOPA and the evidence presented to support this pathway.

6. (Extra credit: 1 point each) Where is B. Hurlburt from? Where is J. Klinman from?